Implementation Cheat Sheet: Lengnick (2013) Macroeconomic Model

Agent-Based Macroeconomics: Baseline Specification, Entity Schemas & Process Dynamics

Authoritative reference: Matthias Lengnick (2013), Journal of Economic Behavior & Organization, Vol. 86, pp. 102–120.

1. Executive Summary & Architectural Overview

Decentralized Macroeconomic Microfoundations

The Lengnick model is a benchmark decentralized, out-of-equilibrium agent-based macroeconomic model featuring **two agent classes** (Households and Firms) without a central Walrasian auctioneer, commercial banks, or government interventions.

PHASE 1: MONTHLY PLANNING

Day 1 Morning (Every 21 Ticks):
- Firms review wages (w_f^{new})
- Labor buffer targets (i_f vs $\underline{\phi}, \overline{\phi}$)
- Calvo sticky price adjustments (p_f)
- Type A supplier rewiring ($n = 7$)
- Labor market matching (w_f vs ω_h)
- Keynesian consumption budget (c_h^r)

PHASE 2: DAILY OPERATIONS

Days 1 to 21 (Daily Trading Loop):
- Randomized household shopping
- Sequential visits across 7 suppliers
- Cash-in-advance purchases (q^*)
- Firm daily linear production ($y_f = \lambda l_f$)
- Inventory accumulation ($i_f += y_f$)
- Unmet demand logged for rewiring

PHASE 3: MONTH-END SETTLEMENT

Day 21 Close (Accounting Reset):
- Contractual wage payments ($W_f = w_f l_f$)
- Aggregate profit clearing (Π_{total})
- Wealth-proportional dividends
- Firm cash reset ($m_f = 0$ strict)
- Reservation wage adaptation (ω_h)
- Sever deferred firings & roll time

Key Architectural Tenets:

1. **Dual Time Horizons**: Daily operational frequency ($t = 1, ..., T_{\text{sim}} \cdot 21$) for goods transactions and physical production; monthly accounting frequency ($T = 21$ business days) for labor contracts, price revisions, and wage disbursements.
2. **Decentralized Bipartite Networks**:
   - **Type A (Commercial Links)**: Directed buyer-seller connections. Each household is linked to exactly $n = 7$ supplier firms ($\text{out-degree} = 7$). Households possess zero global price information.
   - **Type B (Employment Links)**: Directed employer-worker contracts. Each worker has at most 1 employer ($\text{in-degree} \in \{0, 1\}$); each firm employs l_f workers ($\text{out-degree} = l_f$).
3. **Pure Exchange Economy & Strict Money Conservation**: Money functions purely as a medium of exchange. Total money stock M_{total} is strictly invariant across all daily transactions and monthly settlements: $\sum_{h=1}^{H} m_h + \sum_{f=1}^{F} m_f = M_{\text{total}} (\forall t)$. At month end, all firm cash balances are fully distributed as wages and dividends, enforcing $m_f = 0$ at the start of every month.

2. Global Configuration & Parameter Reference

Table 1 (Lengnick 2013, p. 105)

The baseline model is parameterized to calibrate an economy of 1000 workers and 100 firms over a 500-year horizon ($T_{\text{sim}} = 6000$ months) preceded by a 1000-month initialization burn-in:

Config Field Name	Symbol	Scope	Value	Description & Behavioral Function
`num_households`	H	Global	`1000`	Total workforce / household agents. Inelastic labor supply of 1 unit each.
`num_firms`	F	Global	`100`	Total enterprise agents (fixed size; no firm entry/exit in baseline model).
`days_per_month`	T	Global	`21`	Business trading days per macro planning month (1 year = $12 \cdot 21 = 252$ days).
`simulation_horizon_months`	T_{sim}	Global	`6000`	Recorded horizon: 500 years (6000 months) post-burn-in.
`burn_in_months`	$T_{\text{burn-in}}$	Global	`1000`	Initial startup transient periods (months) discarded prior to data collection.
`consumption_exponent`	α	Household	`0.9`	Non-linear Keynesian marginal propensity to consume curvature ($0 < \alpha < 1$).
`supplier_network_size`	n	Household	`7`	Exact out-degree of commercial Type A links per household.
`price_search_prob`	ψ_{Price}	Household	`0.25`	Monthly probability a household seeks a cheaper supplier among larger firms.
`price_discount_threshold`	ξ	Household	`0.01`	Minimum price reduction required (1%) to trigger replacing a commercial link.
`stockout_rewire_prob`	ψ_{Quant}	Household	`0.25`	Monthly probability a quantity-constrained buyer replaces a stock-out supplier.
`unemployed_search_scope`	β	Household	`5`	Maximum number of random firms visited per month by an unemployed jobseeker.
`on_the_job_search_prob`	π	Household	`0.1`	Monthly probability that an employed, satisfied worker initiates on-the-job search.
`reservation_wage_markdown`	$\Delta\omega$	Household	`0.10`	Fractional reduction (10%) applied to reservation wage ω_h after an idle month.
`satisfaction_threshold`	s_{target}	Household	`0.95`	Daily shopping stopping rule: terminates once $\geq 95\%$ of daily quota is bought.
`labor_productivity`	λ	Firm	`3`	Daily goods produced per employed worker: $y_f = \lambda \cdot l_f = 3 \cdot l_f$.
`wage_shock_bound`	δ	Firm	`0.019`	Upper bound of uniform wage adjustment shock: $\mu \sim U[0, \delta]$.

Config Field Name	Symbol	Scope	Value	Description & Behavioral Function
`full_employment_memory`	γ	Firm	24	Consecutive months operating at full staffing required before lowering wage rate.
`inventory_lower_bound`	$\underline{\phi}$	Firm	0.25	Lower inventory buffer: $\underline{i}_f = \underline{\phi} \cdot d_f^{\text{old}}$ (triggers hiring / price hikes).
`inventory_upper_bound`	$\overline{\phi}$	Firm	1.0	Upper inventory buffer: $\overline{i}_f = \overline{\phi} \cdot d_f^{\text{old}}$ (triggers firings / price cuts).
`price_markup_min`	$\underline{\varphi}$	Firm	1.025	Lower bound on goods price relative to marginal cost: $\underline{p}_f = \underline{\varphi} \cdot \text{mc}_f$.
`price_markup_max`	$\overline{\varphi}$	Firm	1.15	Upper bound on goods price relative to marginal cost: $\overline{p}_f = \overline{\varphi} \cdot \text{mc}_f$.
`price_adjustment_bound`	ϑ	Firm	0.02	Upper bound of uniform price adjustment step: $\nu \sim U[0, \vartheta]$.
`calvo_price_prob`	θ	Firm	0.75	Calvo price friction: probability of successfully implementing a desired price change.
`total_money_stock`	M_{total}	Global	100000	Invariant nominal money stock endowed into the banking-free closed economy.
`initial_household_liquidity`	$m_{h,0}$	Household	100.0	Uniform starting cash endowed to each household ($\frac{M_{\text{total}}}{H}$).
`initial_firm_liquidity`	$m_{f,0}$	Firm	0.0	Starting cash balance for all firms ($t = 0$); cash accumulated solely from sales.
`initial_wage`	$w_{f,0}$	Firm	1.0	Uniform baseline monthly contractual wage rate offered by all firms at $t = 0$.
`initial_reservation_wage`	$\omega_{h,0}$	Household	1.0	Uniform initial reservation wage ($\omega_{h,0} = w_{f,0}$ ensures initial satisfaction).
`initial_workers_per_firm`	$l_{f,0}$	Firm	10	Initial labor force per firm ($\frac{H}{F} = 10$), establishing 0% starting unemployment.
`initial_price_markup`	φ_{init}	Firm	0.10	Initial markup (10%) above marginal cost: $p_{f,0} = 1.10 \cdot \text{mc}_{f,0} \approx 0.01746$.
`initial_inventory`	$i_{f,0}$	Firm	630	Initial warehouse stock: exactly 1 month of normal production $(21 \cdot \lambda \cdot l_{f,0})$.
`initial_demand_history`	$d_{f,0}^{\text{old}}$	Firm	630	Baseline monthly demand history $(21 \cdot \lambda \cdot l_{f,0})$ anchoring buffer targets.
`initial_consecutive_full`	$\text{cfm}_{f,0}$	Firm	24	Initial full-employment counter $(\gamma = 24)$, allowing wage cuts if idle.

3. Entity Tables & Schema Specifications

Data Types, Invariants & Foreign Keys

3.1 Household Entity Schema (H = 1000)

Each household inelastically supplies 1 unit of labor, receives income (wages and profit shares), and purchases consumption goods:

Field Name	Data Type	Constraint / Default	Role & Behavioral Meaning
`id`	`int`	Primary Key: $1 \ldots H$	Unique household identifier.
`liquidity` (m_h)	`float`	$m_h \geq 0$	Cash holdings. Bounded by cash-in-advance; cannot be overdrawn.
`reservation_wage` (ω_h)	`float`	$\omega_h > 0$	Minimum acceptable wage for employment. Ratchets on wage hike; drops 10% when idle.
`employer_id`	`int | null`	Foreign Key $\rightarrow$ Firm	Type B connection. Current employer (`null` if worker is currently unemployed).
`supplier_ids`	`List[int]`	Size $\lvert\text{supplier\_ids}\rvert = 7$	Type A connections to 7 supplier firms. Directed buyer-seller network.
`monthly_real_budget` (c_h^r)	`float`	$c_h^r \geq 0$	Planned real consumption volume for current month: $\min\left\{\left(\frac{m_h}{P_h^I}\right)^\alpha, \frac{m_h}{P_h^I}\right\}$.
`daily_real_demand` (c_h^{daily})	`float`	$= \frac{c_h^r}{21}$	Daily consumption demand quota target evaluated each morning.
`constrained_firms`	`Dict[int, float]`	Default: `{}`	Maps supplier firm ID to unsatisfied real demand volume caused by inventory stockouts.
`is_employed`	`bool`	Derived (`employer != null`)	Convenience status flag indicating employment status $(\text{in-degree} == 1)$.
`is_satisfied`	`bool`	Derived ($w_{\text{empl}} \geq \omega_h$)	Job satisfaction indicator. Satisfied workers search with $\pi = 0.1$; unsatisfied search with 1.0.

3.2 Firm Entity Schema (F = 100)

Firms employ workers, produce consumption goods with linear technology, manage inventories, set prices and wages, and disburse profits:

Field Name	Data Type	Constraint / Default	Role & Behavioral Meaning
`id`	`int`	Primary Key: $1 \ldots F$	Unique enterprise identifier.
`liquidity` (m_f)	`float`	Signed float	Operational cash balance. Accumulates sales revenue; deducted for payroll; cleared to 0 monthly.
`inventory` (i_f)	`float`	$i_f \geq 0$	Stock of finished goods ready for customer dispatch. Increased by production; reduced by sales.

Field Name	Data Type	Constraint / Default	Role & Behavioral Meaning
`price` (p_f)	`float`	$p_f \in [\underline{p}_f, \overline{p}_f]$	Current selling price per unit. Sticky adjustment via Calvo parameter $\theta = 0.75$.
`wage` (w_f)	`float`	$w_f > 0$	Monthly wage offered to all existing workers and new recruits.
`employees`	`Set[int]`	Foreign Keys $\to$ Household	Set of active employee IDs (size $l_f = \lvert\text{employees}\rvert$). Determines daily production batch.
`vacancies`	`int`	$\in \{0, 1\}$	Number of open positions offered this month. Set to 1 if $i_f < \underline{i}_f$, else 0.
`unfilled_vacancy_last_month`	`bool`	Default: `false`	Flag indicating whether firm posted a vacancy last month that failed to attract a hire.
`consecutive_full_months`	`int`	≥ 0	Consecutive months with all positions filled. Triggers wage reduction when ≥ 24.
`pending_firings`	`List[int]`	Default: `[]`	Employees marked for dismissal at month end due to job protection statutory lag.
`demand_last_month` (d_f^{old})	`float`	$d_f^{\text{old}} \geq 0$	Total units demanded in previous month. Anchors inventory buffer thresholds $\underline{i}_f$ and $\overline{i}_f$.
`demand_current_month`	`float`	Accumulator (≥ 0)	Running total of real demand intentions received across all 21 trading days.
`marginal_cost` (mc_f)	`float`	Derived: $\frac{w_f}{21 \cdot \lambda}$	Unit labor cost of production: $\text{mc}_f = \frac{w_f}{63}$. Bounds pricing boundaries.

3.3 Network Topology (Relational Bipartite View)

The macroeconomy is governed by two directed bipartite graphs linking Households and Firms:
- **Type A Links (Goods Market Network)**: Directed connection from Household h to Firm f. Out-degree of every household is strictly fixed at $n = 7$. In-degree of firm f is unconstrained ($\in [0, H]$). Captures brand loyalty, geographic separation, and localized information.
- **Type B Links (Labor Market Network)**: Directed contract from Firm f to Household h. In-degree of household is at most 1 (0 if unemployed). Out-degree of firm is its workforce size l_f.

4. Process Dynamics: Step-by-Step Algorithmic Logic

Sequential Execution Specifications

> **⚡ Execution Ordering Policy**
>
> The simulation loop must execute in strict chronological sequence. Random agent orderings must be freshly reshuffled before each loop pass to eliminate structural primacy bias.

Phase 1: Beginning-of-Month Operations (Planning & Network Matching)

Executed on the morning of Day 1 ($t \equiv 1 \bmod 21$) before any market transactions.

Step 1.1: Firm Wage Review & Adaptation (`PROC-1.1`) Each firm f evaluates its past staffing conditions:
- **Labor Shortage (Wage Hike)**: If firm had an unfilled vacancy last month (`unfilled_vacancy_last_month == true`):

$w_f^{\text{new}} = w_f^{\text{old}} \cdot (1 + \mu)$, with uniform wage shock $\mu \sim U[0, \delta]$ ($\delta = 0.019$). Reset counter: `consecutive_full_months = 0`.
- **Labor Surplus (Wage Cut)**: Else if firm remained fully staffed for at least $\gamma = 24$ consecutive months (`consecutive_full_months >= 24`):

$w_f^{\text{new}} = w_f^{\text{old}} \cdot (1 - \mu)$, with uniform wage shock $\mu \sim U[0, \delta]$.
- **Neutral**: Otherwise, preserve existing wage rate: $w_f^{\text{new}} = w_f^{\text{old}}$.

Step 1.2: Firm Production Planning & Labor Targets (`PROC-1.2`) Firms evaluate inventory i_f against buffer thresholds derived from previous month demand d_f^{old}:

$$\underline{i}_f = \underline{\phi} \cdot d_f^{\text{old}} = 0.25 \cdot d_f^{\text{old}}, \qquad \overline{i}_f = \overline{\phi} \cdot d_f^{\text{old}} = 1.0 \cdot d_f^{\text{old}}$$

- **Case 1 (Inventory Deficit, $i_f < \underline{i}_f$)**: Open 1 vacancy: `vacancies = 1`.
- **Case 2 (Inventory Excess, $i_f > \overline{i}_f$ and $l_f > 0$)**: Select random worker $h \in \text{employees}$; schedule in `pending_firings` (dismissal executes after 1-month statutory job-protection lag).
- **Case 3 (Normal Buffer, $\underline{i}_f \leq i_f \leq \overline{i}_f$)**: No hiring or firing; `vacancies = 0`.

Step 1.3: Firm Calvo Sticky Price Setting (`PROC-1.3`) Prices adjust only when inventory breaches buffer thresholds:
- Nominal marginal cost: $\text{mc}_f = \frac{w_f}{21 \cdot \lambda} = \frac{w_f}{63}$.
- Markup bounds: $\underline{p}_f = \underline{\varphi} \cdot \text{mc}_f = 1.025 \cdot \text{mc}_f$, and $\overline{p}_f = \overline{\varphi} \cdot \text{mc}_f = 1.15 \cdot \text{mc}_f$.
- **Price Hike ($i_f < \underline{i}_f$)**: If $p_f < \overline{p}_f$, with probability $\theta = 0.75$: draw $\nu \sim U[0, \vartheta]$ ($\vartheta = 0.02$) and set $p_f^{\text{new}} = p_f^{\text{old}} \cdot (1 + \nu)$.
- **Price Cut ($i_f > \overline{i}_f$)**: If $p_f > \underline{p}_f$, with probability $\theta = 0.75$: draw $\nu \sim U[0, \vartheta]$ and set $p_f^{\text{new}} = p_f^{\text{old}} \cdot (1 - \nu)$.
- **Sticky Default**: With probability $1 - \theta = 0.25$, or inside buffer, price remains unchanged.

Step 1.4: Consumer Network Rewiring (`PROC-1.4`) Households adapt their 7-supplier commercial network:
1. **Price Search (Prob $\psi_{\text{Price}} = 0.25$)**:
   - Pick connected firm $f_{\text{curr}} \in \text{supplier\_ids}_h$ uniformly at random.
   - Sample candidate firm $f_{\text{cand}} \notin \text{supplier\_ids}_h$ with probability proportional to workforce size: $P(f_{\text{cand}} = k) \propto l_k$.
   - If $p_{f_{\text{cand}}} \leq (1 - \xi) \cdot p_{f_{\text{curr}}}$ (at least $\xi = 1\%$ cheaper): swap link ($\text{remove } f_{\text{curr}}, \text{insert } f_{\text{cand}}$).
2. **Quantity-Constraint Search (Prob $\psi_{\text{Quant}} = 0.25$)**:
   - If `constrained_firms` is non-empty: select f_{bad} with probability proportional to unmet demand magnitude.

- Pick unconnected firm f_{new} uniformly at random; swap link ($\text{remove } f_{\text{bad}}, \text{insert } f_{\text{new}}$).

Step 1.5: Labor Market Search & Matching (`PROC-1.5`) Process households in randomized sequence across three search tiers:
- **Tier 1 (Unemployed Workers)**: Visit up to $\beta = 5$ random firms sequentially. If visited firm has `vacancies > 0` and $w_f \geq \omega_h$: worker hired! Form Type B link, decrement f's vacancies by 1, and terminate search.
- **Tier 2 (Employed & Unsatisfied, $w_{\text{current}} < \omega_h$)**: Search with probability 1.0. Visit 1 random firm f. If f has `vacancies > 0` and $w_f > w_{\text{current}}$: switch employer (sever old Type B link, form new Type B link, decrement f's vacancies by 1).
- **Tier 3 (Employed & Satisfied, $w_{\text{current}} \geq \omega_h$)**: Search with probability $\pi = 0.1$. Visit 1 random firm f. If f has `vacancies > 0` and $w_f > w_{\text{current}}$: switch employer.

Step 1.6: Household Consumption Budget Planning (`PROC-1.6`) Households calculate local price average across their 7 suppliers:

$$P_h^I = \frac{1}{7} \sum_{f \in \text{supplier\_ids}_h} p_f$$

Planned monthly real consumption follows Keynesian concave demand bounded by cash wealth:

$$c_h^r = \min\left\{ \left(\frac{m_h}{P_h^I}\right)^{\alpha}, \frac{m_h}{P_h^I} \right\}, \quad \alpha = 0.9$$

Daily real consumption quota: $c_h^{\text{daily}} = \frac{c_h^r}{21}$.

Phase 2: Daily Operations Loop (Days 1 to 21)

Executed every business day $d \in \{1, ..., 21\}$ in strict sequence: Consumption Market → Production.

Step 2.1: Consumption Market Transactions (`PROC-2.1`)
1. Reshuffle all households into a randomized sequence.
2. For each household h:
   - Set remaining quota: $q_{\text{rem}} = c_h^{\text{daily}}$; visited counter: $k = 0$.
   - While $k < n$ ($k < 7$) and $q_{\text{rem}} > 0.05 \cdot c_h^{\text{daily}}$ (until $\geq 95\%$ fulfilled):
     - Pick unvisited supplier $f \in \text{supplier\_ids}_h$; increment $k \to k + 1$.
     - Max affordable units: $q_{\text{aff}} = \frac{m_h}{p_f}$; Available units: $q_{\text{avail}} = i_f$.
     - Desired units: $q_{\text{des}} = \min(q_{\text{rem}}, q_{\text{aff}})$.
     - Executed trade quantity:

   $q^* = \min(q_{\text{des}}, q_{\text{avail}})$.
     - Log demand: $f.\text{demand\_current\_month} += q_{\text{des}}$.
     - Log stockout: If $q^* < q_{\text{des}}$, record unmet $(q_{\text{des}} - q^*)$ in $h.\text{constrained\_firms}[f]$.
     - Settle trade: $m_h \to m_h - q^* p_f$, $m_f \to m_f + q^* p_f$, $i_f \to i_f - q^*$, $q_{\text{rem}} \to q_{\text{rem}} - q^*$.
   - **Any unmet demand at day end vanishes without backlog.**

Step 2.2: Daily Production Process (`PROC-2.2`) For every firm $f \in \{1, ..., F\}$ on day d:
- Daily manufactured output: $y_f = \lambda \cdot l_f = 3 \cdot l_f$.
- Inventory accumulation: $i_f \to i_f + y_f$.
- Daily Aggregate GDP: $Y_d = \sum_{f=1}^{F} y_f$.

Phase 3: Month-End Settlement & Revision (Day 21 Night)

Executed at the conclusion of day 21. Handles accounting settlements, cash reset, and reservation wages.

Step 3.1: Payroll & Wage Disbursement (`PROC-3.1`) For each firm f:
- Total payroll obligation: $W_f = w_f \cdot l_f$.
- Deduct wage bill: $m_f \to m_f - W_f$ **(Note: m_f may turn negative if monthly sales revenue was below payroll).**
- For each employee $h \in \text{employees}_f$: transfer wage: $m_h \to m_h + w_f$.

Step 3.2: Profit Redistribution & Strict Firm Cash Reset (`PROC-3.2`)
- Aggregate operating balance: $\Pi_{\text{total}} = \sum_{f=1}^{F} m_f$.
- Aggregate household liquidity: $m_{\text{total}}^{HH} = \sum_{h=1}^{H} m_h$.
- Distribute dividend/deficit shares proportional to cash holdings:

$\Delta m_h = \Pi_{\text{total}} \cdot \left(\frac{m_h}{m_{\text{total}}^{HH}}\right)$, then update $m_h \to m_h + \Delta m_h$.
- **Strict Firm Cash Reset**: Reset every firm cash balance identically to zero:

$m_f \to 0.0$ for all $f \in \{1, ..., F\}$.

Step 3.3: Household Reservation Wage Update (`PROC-3.3`)
- **Employed Worker**: If wage earned $w_{\text{rec}} > \omega_h$, ratchet upward: $\omega_h \to w_{\text{rec}}$. If $w_{\text{rec}} \leq \omega_h$, keep ω_h unchanged (worker becomes unsatisfied; searches with probability 1.0 next month).
- **Unemployed Worker**: Apply statutory 10% markdown: $\omega_h \to 0.90 \cdot \omega_h$.

Step 3.4: Administrative Carryover & Deferred Firings (`PROC-3.4`)
1. **Execute Firings**: For each worker $h \in \text{pending\_firings}_f$, sever Type B link (`employer_id = null`, remove from firm's employees). Clear `pending_firings = []`.
2. **Staffing Continuity**: If `vacancies > 0`, set `unfilled_vacancy = true`, `consecutive_full = 0`. Else if `vacancies == 0` and $l_f > 0$, set `unfilled_vacancy = false`, `consecutive_full += 1`. Reset `vacancies = 0`.
3. **Demand Rollover**: Copy $d_f^{\text{old}} \to \text{demand\_current\_month}$, then reset `demand_current_month = 0`.
4. **Clear Stockout Logs**: Empty `constrained_firms = {}` for all households.

5. Initialization State & Simulation Invariants

Defensive Programming & Verification Checks

5.1 Initialization Checklist (Equilibrium Start)

To eliminate artificial simulation startup transients, initialize the state close to stationary levels:

1. **Total Money Supply**: Set $M_{\text{total}} = 100,000$. Allocate $m_h = \frac{M_{\text{total}}}{H} = 100.0$ to each household; set all firm cash $m_f = 0.0$.
2. **Initial Employment**: Assign exactly 10 workers to each of the 100 firms ($\frac{1000}{100} = 10$). Initial unemployment is 0%.
3. **Initial Wages & Reservations**: Set $w_f = 1.0$ for all firms; set $\omega_h = 1.0$ for all households.
4. **Marginal Cost & Prices**: Marginal cost is $\text{mc}_f = \frac{1.0}{21 \cdot 3} \approx 0.01587$. Set initial prices to a 10% markup: $p_f = 1.10 \cdot \text{mc}_f \approx 0.01746$.
5. **Initial Inventories & Demand History**: Pre-fill warehouses with 1 month of normal production:

$$i_{f,0} = 21 \cdot \lambda \cdot l_f = 21 \cdot 3 \cdot 10 = 630\,\text{units}, \qquad d_{f,0}^{\text{old}} = 630\,\text{units}$$

6. **Initial Network Graphs**: Draw $n = 7$ distinct supplier IDs uniformly at random for each household.

5.2 Strict Simulation Invariant Assertions

Embed these 6 mathematical invariant checks directly into your simulation engine loop:

✔ Simulation Invariant Assertions

1. Global Money Conservation: $\lvert\sum_{h=1}^{H} m_h + \sum_{f=1}^{F} m_f - M_{\text{total}}\rvert < 10^{-7}$	Verify at every daily and monthly tick t. Guarantees zero cash leakage during transactions, payroll, or dividend distribution.
2. Firm Cash Solvency: $m_f == 0.0 \quad \forall f \in \{1, ..., F\}$	Verify on Day 1 morning of each month. All operating cash must be strictly cleared via `PROC-3.2`.
3. Household Budget Solvency: $m_h \geq 0.0 \quad \forall h \in \{1, ..., H\}$	Verify before each purchase. Overdrafts strictly prohibited; trade bounded by cash-in-advance ($q^* \leq \frac{m_h}{p_f}$).
4. Non-Negative Inventory: $i_f \geq 0.0 \quad \forall f \in \{1, ..., F\}$	Verify after daily trades. Stockout enforcement prevents warehouse inventory from dipping below zero.
5. Labor Accounting Invariant: $\sum_{f=1}^{F} l_f + \lvert\{h : \text{employer}_h == \text{null}\}\rvert == H = 1000$	Verify after matching (`PROC-1.5`) and firings (`PROC-3.4`). No agent may vanish from labor register.
6. Network Degree Preservation: $\lvert\text{supplier\_ids}_h\rvert == 7 \quad \forall h \in \{1, ..., H\}$	Verify after rewiring (`PROC-1.4`). Commercial network strictly preserves regular out-degree $n = 7$.

6. Implementation Verification & Stylized Facts

Model Validation Signatures (Lengnick 2013, Sec 3–5)

When your simulation is bug-free and executing properly, aggregate macroeconomic time-series will spontaneously reproduce the empirical stylized facts documented by Lengnick:

Empirical Stylized Fact	Expected Quantitative Signature in Model	Paper Citation
Spontaneous Coordination	Aggregate employment rate hovers near full employment: $95.8\%...100\%$ (unemployment $u \in [0\%, 4.2\%]$), demonstrating decentralized out-of-equilibrium stability.	Sec 3, Fig 6
Excess Demand Rarity	Unsatisfied consumption demand is $< 0.65\%$ in over 95% of all simulated months; shortages remain highly localized without systemic cascading failure.	Sec 3, Fig 5
Beveridge Curve	Robust negative empirical correlation between vacancy rate and unemployment rate (u vs v trade-off).	Sec 3, Fig 7b
Wage Phillips Curve	Negative correlation between unemployment rate and nominal wage growth / goods price inflation.	Sec 3, Fig 7a
Right-Skewed Firm Sizes	Firm workforce distribution is heavily right-skewed with sample skewness ≈ 1.72 and excess kurtosis ≈ 3.73, matching empirical Zipf-type firm size distributions.	Sec 3, Fig 8
Calvo Price Stickiness	Median frequency of price changes across firms is $\approx 11\%$ per month, matching empirical microeconomic studies of sticky consumer prices.	Sec 3
GDP-Inflation Lag Dynamics	Real GDP positively correlates with inflation with a 2-quarter lag (peaking around 6 months), consistent with empirical VAR macro literature (Walsh 2003).	Sec 3, Fig 9
Long-Run Money Neutrality	Doubling nominal money stock M_{total} leads to exactly doubled price levels along a $45°$ ray with zero long-run effect on aggregate real output or employment.	Sec 4, Fig 11
Short-Run Money Non-Neutrality	A 5% positive money shock creates an immediate cyclical boom in real GDP peaking in year 1, decaying smoothly over 2 to 3 years before returning to baseline.	Sec 4, Fig 12
Endogenous Butterfly Effect	Perturbing just 1 household's demand by 5% on a single day causes the trajectory of aggregate GDP to fully diverge from baseline after several decades.	Sec 5, Fig 13

Lengnick Baseline Economic Model Implementation Reference Cheat Sheet — Compiled with Typst 0.15.1